

Thermal Conductivity Prediction at 300K

Akshat Mittal Harshit Dashore

Supervisor: Prof. Somnath Bhowmick

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Project Objective

- Predict thermal conductivity at 300K (TC_300) from material descriptors.
- Build a leakage-safe, interpretable ML pipeline with reproducible grouped-CV metrics.
- Present final submission through Regression, Classification, and PCA only.



Work Sequence (End-to-End)

- 1 Internal data loading, EDA, duplicate cleanup, and descriptor pruning.
- 2 Track-A: internal-only regression benchmark after the final drop policy.
- 3 Track-B: matminer-only regression benchmark, feature-count tuning, and interpretation.
- 4 Track-A classification retained as a separate auxiliary block.
- 5 PCA retained as a separate diagnostic block for Track-A and Track-B.



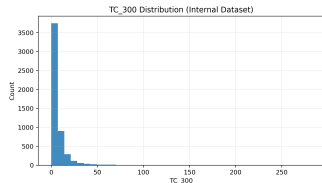
Track Definition (Strict Separation)

- **Track-A (internal-only):** only internal descriptors from `thermal_data.xlsx`.
- **Track-B (matminer-only):** only `mm_` composition descriptors (132 total).
- No mixing of internal and matminer features inside the primary regression tables.
- Regression, Classification, and PCA are shown in separate sections for a cleaner final submission flow.



Original Dataset and EDA

- Source: `thermal_data.xlsx`
- Raw rows before policy: 5,454
- Rows after formula + SGN deduplication: 5,263
- Duplicate rows removed: 191
- Final Track-A regression uses 13 internal numeric descriptors; Track-B uses 132 matminer descriptors.



Interpretation: TC₃₀₀ is right-skewed, so RMSE and MAPE do not always move together. This is also why the regression target is modeled with a log transform and then converted back for final reporting.

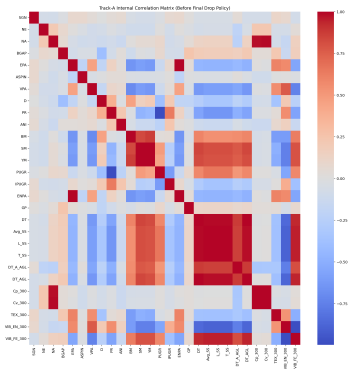


Preprocessing and Standardization Policy

- Numeric preprocessing: median imputation inside all pipelines.
- Standardization used for linear and PCA-style blocks.
- Tree ensembles are run without mandatory scaling.
- Regression target is trained as $\log_{10}(\text{TC}_{300})$ and inverse-transformed for final reporting.
- The next slides report **Track-A results** using internal descriptors only.



Feature Removal (Final Drop List)



Final removed descriptors:

TEX_300, YM, T_SS, Avg_SS, L_SS,
NA, DT_AGL, EPA, VIB_EN_300,
VIB_FE_300, PR, Cv_300, NE, BGAP,
ASPIN

Track-A after pruning:

- 15 features are removed.

Interpretation: Some features are dropped because of high correlation, whereas others are removed because they are not useful for predicting thermal conductivity.



Internal Baseline Recheck

Model	R^2	MAE	RMSE	MAPE (%)
Extra Trees	0.8657	1.1289	4.9302	14.93
Random Forest	0.8617	1.1909	5.0107	15.60
Gradient Boosting	0.8607	1.2169	5.0224	16.57
RidgeCV	0.8149	1.6943	5.7832	25.41
Linear Regression	0.8133	1.6951	5.8041	25.36
ElasticNetCV	0.8128	1.6988	5.8131	25.46

Interpretation: Track-A remains the strongest pure-track regression block after the final drop policy.



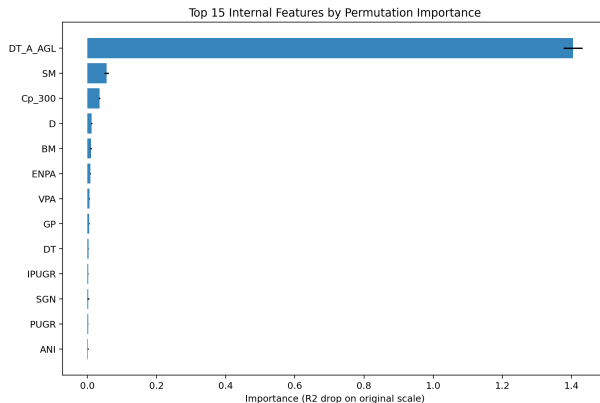
Stacked Ensemble Result

Model	R^2	MAE	RMSE	MAPE (%)
Stacking Regressor (RF + ET + GB -> RidgeCV)	0.8660	1.1051	4.9183	13.99

- Feature set used: **Track-A internal descriptors only (13 features)**.
- Base learners: Random Forest, Extra Trees, Gradient Boosting.
- Meta learner: RidgeCV trained on the full out-of-fold prediction matrix.
- It is reported separately as an ensemble reference, not as the main Track-A vs Track-B comparison table.



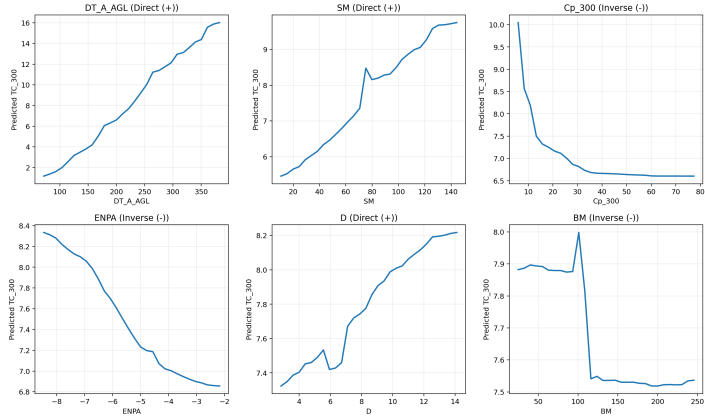
Top Feature Importance



Interpretation: Track-A importance remains concentrated in a small internal descriptor set. In the executed notebook, DT_A_AGL, SM, Cp_300, D, BM, and ENPA are the main contributors.



PDP Curves (Direct/Inverse Relationship)



Interpretation: The Track-A PDP block supports directional interpretation: DT_A_AGL, SM, and D show direct trends, while Cp₃₀₀, ENPA, and BM show inverse behaviour.

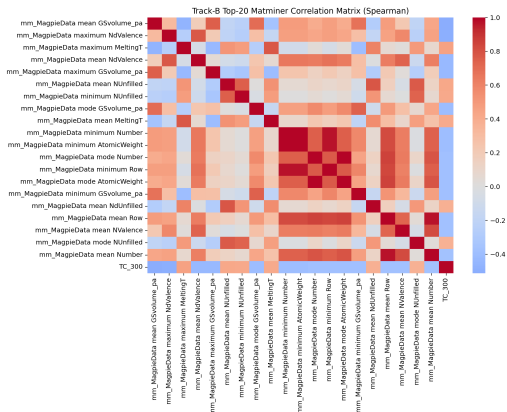


Track-B Setup: Matminer Composition Features

- Track-B uses `matminer` composition features generated from formula only.
- These descriptors summarize composition statistics such as valence, atomic weight, melting temperature, row/group trends, and related Magpie properties.
- Track-B is intentionally composition-only, so it does not use the internal structural descriptors available in Track-A.
- The goal is to test how far composition-based features alone can predict TC₃₀₀, and then tune a reduced- k version of the same feature family.



Track-B Correlation (Matminer-Only)



Interpretation: This top-20 Spearman heatmap is used first to inspect how the strongest composition descriptors relate to each other and to TC_300. It helps explain redundancy and descriptor families before the Track-B regression tables are compared.



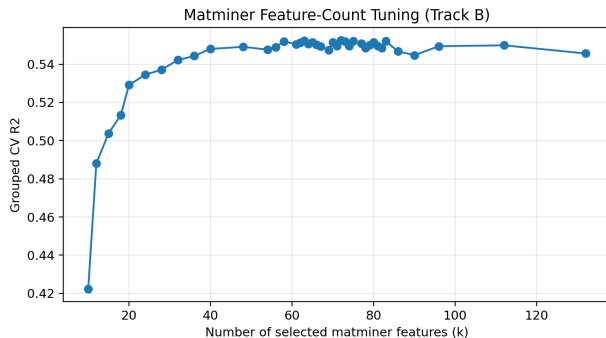
Track-B Full-132 Leaderboard (Executed 6 Models)

Model	R^2	MAE	RMSE	MAPE (%)
GradientBoosting_MM	0.5854	3.2680	8.7082	69.74
RandomForest_MM	0.5844	3.0916	8.6746	71.90
ExtraTrees_MM	0.5478	3.1308	8.9779	71.06
LinearRegression_MM	0.4867	3.7866	9.7051	94.86
RidgeCV_MM	0.4857	3.7905	9.7163	94.95
ElasticNetCV_MM	0.4840	3.7979	9.7345	95.17

Interpretation: Track-B uses only composition-based matminer descriptors. It captures broad composition trends, but it remains weaker than Track-A because structural/internal information is not present in this feature family.



Track-B Feature-Count Tuning



- Start with all 132 matminer descriptors.
- Use `SelectKBest(mutual_info_regression)` to rank descriptors for each candidate k .
- Evaluate candidate k values under grouped CV with an Extra Trees tuning model.
- Choose the best k from the grouped-CV tuning table, then rerun the reduced Track-B model family on that selected feature set.
- Final reduced Track-B export uses $k = 72$.



Track-B Reduced- k Results ($k = 72$)

Model	R^2	MAE	RMSE	MAPE (%)
RandomForest_MM_reduced	0.5845	3.0846	8.6680	71.03
GradientBoosting_MM_reduced	0.5811	3.2823	8.7563	71.27
ExtraTrees_MM_reduced	0.5509	3.1070	8.9510	69.98
RidgeCV_MM_reduced	0.4653	3.8492	9.9079	108.20
ElasticNetCV_MM_reduced	0.4595	3.8647	9.9600	106.62

Interpretation: After grouped-CV tuning selects $k = 72$, the reduced Random Forest model gives nearly the same R^2 as the full 132-feature Track-B leaders and a slightly lower RMSE.



Track-B Selected Features and Reduction Summary

Track-B grouped-CV summary:

- Best full-132 model: GradientBoosting_MM, $R^2 = 0.5854$, RMSE = 8.7082.
- Final reduced-feature count: $k = 72$ from grouped-CV tuning.
- Best reduced model: RandomForest_MM_reduced, $R^2 = 0.5845$, RMSE = 8.6680.
- Keep both views: full-132 gives the best R^2 ; reduced- k is slightly better in RMSE and easier to interpret.

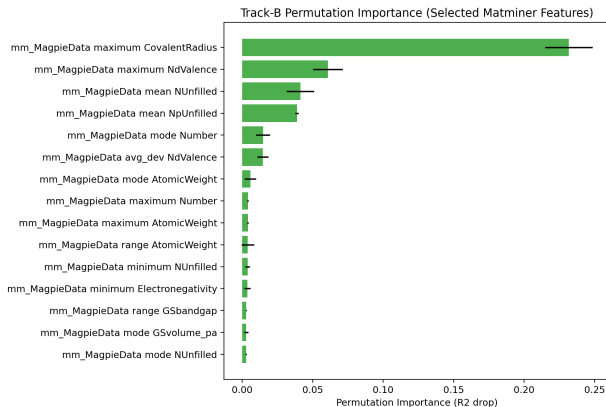
Strong Track-B association examples:

- mm_MagpieData mean GSvolume_pa (inverse)
- mm_MagpieData maximum NdValence (inverse)
- mm_MagpieData maximum MeltingT (direct)
- mm_MagpieData mean NUnfilled (direct)

Interpretation: Track-B captures periodic, valence, and melting-related composition trends. Reduced- k is the grouped-CV tuned compact version of Track-B.



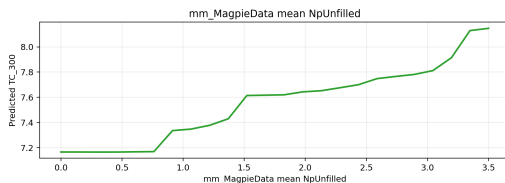
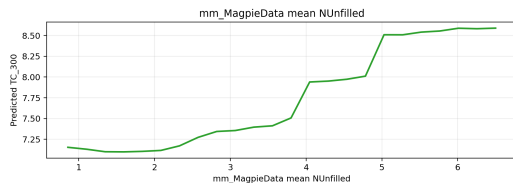
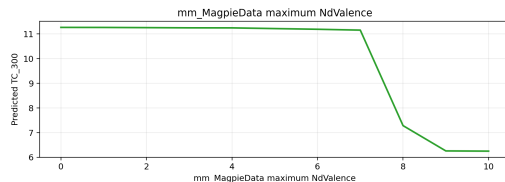
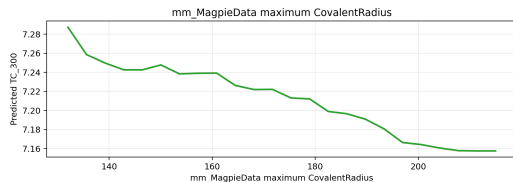
Track-B Permutation Importance (Matminer-Only)



Interpretation: The executed Track-B importance block is led by Nd-valence, GS-volume, number, and atomic-weight statistics. These are interpretation supports, while the main model choice still comes from grouped-CV tables.



Track-B PDP Curves (Matminer-Only)



Interpretation: Track-B PDP curves are reported as directional diagnostics. In the executed run, maximum covalent radius and maximum Nd valence trend downward, while mean NUnfilled and mean NpUnfilled trend upward.



Track-A vs Track-B (Model Parity View)

Model	Track-A R^2	Track-B R^2
Gradient Boosting	0.8607	0.5854
Random Forest	0.8617	0.5844
Extra Trees	0.8657	0.5478
RidgeCV	0.8149	0.4857
ElasticNetCV	0.8128	0.4840
Linear Regression	0.8133	0.4867

Interpretation: Track-A and Track-B were evaluated under the same grouped-CV philosophy with strict feature-family separation. The gap shows that composition-only descriptors do not fully recover the internal signal retained in Track-A.



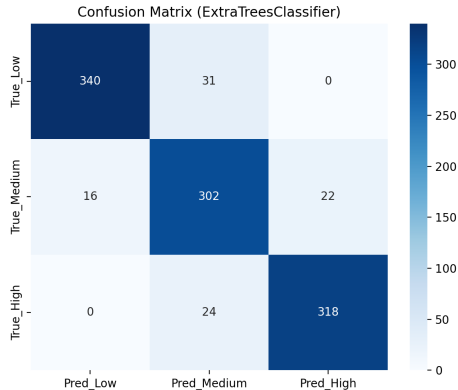
Track-A Classification (Auxiliary Diagnostic)

Classifier	Accuracy	Macro-F1
Extra Trees Classifier	0.9105	0.9112
Random Forest Classifier	0.9103	0.9109
Gradient Boosting Classifier	0.9094	0.9101
Logistic Regression	0.8964	0.8972

Interpretation: This classification block is done on **Track-A internal features only**. A separate Track-B classification run is not included in the notebook because the final matminer-only analysis was reserved for regression, interpretation, and PCA diagnostics.



Track-A Confusion Matrix (Best Classifier)



Interpretation: The best classifier is Extra Trees on Track-A internal features. Most misclassification remains near class boundaries, while extremes are better separated.

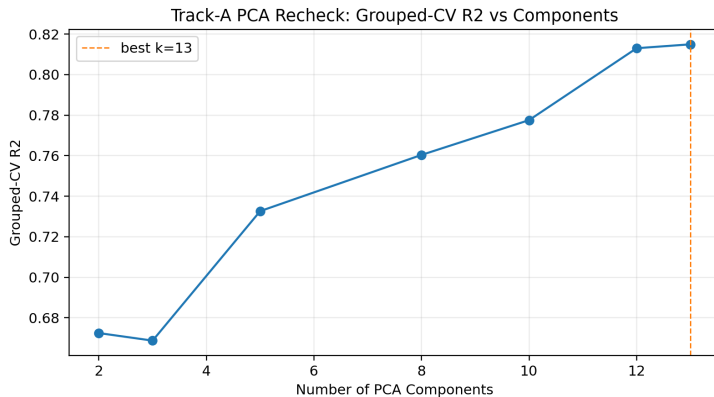


PCA Diagnostic Status

- **Track-A PCA:** best result occurs at 13 components, which equals the full Track-A descriptor count after pruning.
- **Track-B PCA:** 95% explained variance is reached at 34 components and 99% at 57 components.
- However, the best predictive Track-B PCA result still occurs at the full 132-component limit, not at compressed dimensionality.
- Therefore PCA is reported as a diagnostic representation check, not as the final recommended route for either track.



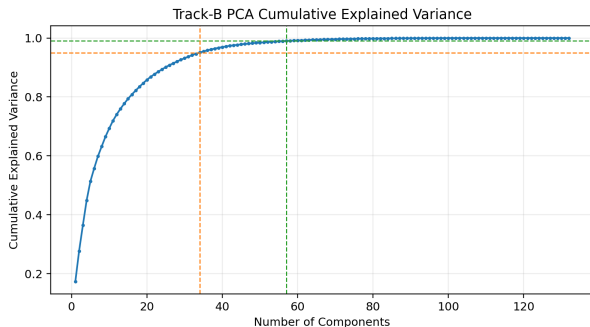
Track-A PCA Component Sweep



Interpretation: Track-A performance improves as components are restored and peaks at the full 13-component limit. This means PCA compression does not improve over the cleaned Track-A descriptor space.



Track-B PCA Status (Executed Run)



PCA quantity	Value
k_{95} explained variance	34
k_{99} explained variance	57
Best PCA R^2	0.4857
Best PCA RMSE	9.7163
Best PCA MAE	3.7905
Best predictive PCA at	132 comps

Interpretation: Track-B PCA is useful as a compression audit for the 132 matminer descriptors. The best predictive PCA result occurs at the full 132-component limit, so PCA remains a diagnostic block only.



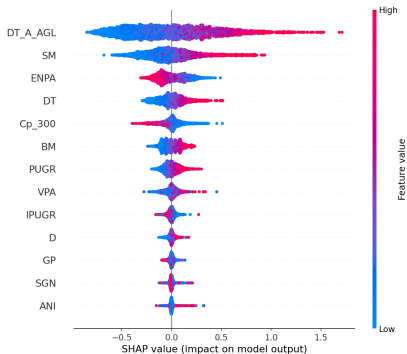
Extension Roadmap Completed

- **Task 1:** SHAP interpretation for the Track-A Extra Trees model.
- **Task 2:** XGBoost optimized with Optuna under SGN-grouped CV.
- **Task 3:** Slack formula validation and physics-informed residual hybrid.
- **Task 4:** CGCNN trained on Materials Project matched crystal structures.
- **Task 5:** Clean final notebook, README, final plots, and reproducible scripts.

Interpretation: The extension work tests three complementary explanations: descriptor ML, analytical physics, and crystal-graph learning.



SHAP: DT_A_AGL Dominates Track-A

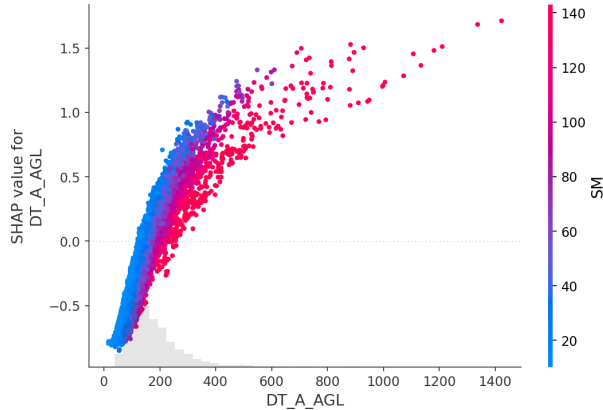


- DT_A_AGL ranks **1st** by mean absolute SHAP value.
- DT_A_AGL also ranks **1st** by permutation importance.
- This confirms Debye temperature is the dominant learned descriptor.

Interpretation: The model's strongest learned signal is the descriptor that appears directly in Slack phonon theory.



SHAP Dependence: Debye Temperature and Shear Modulus

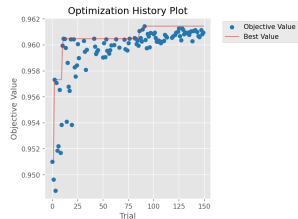


Interpretation: The dependence plot shows how DT_A_AGL changes the Extra Trees prediction, with SM used as the interaction/color variable.



XGBoost + Optuna Result

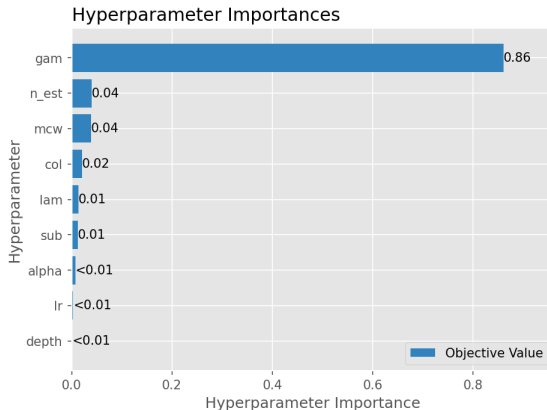
Model	R^2	RMSE	MAPE (%)
Extra Trees baseline	0.8657	4.93	14.93
Random Forest	0.8617	5.01	15.60
Gradient Boosting	0.8607	5.02	16.57
XGBoost + Optuna	0.8995	4.17	16.14



Interpretation: XGBoost was optimized on the same 13 Track-A descriptors using SGN-grouped folds and log-transformed targets.



Optuna Parameter Importance

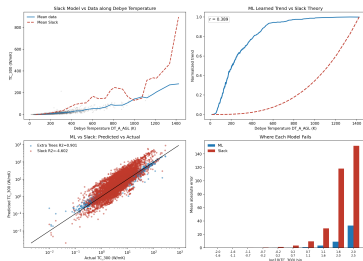


Interpretation: The search diagnostic shows which hyperparameters most influenced XGBoost grouped-CV performance.



Slack Formula Physics Validation

Model	R^2	RMSE	MAPE (%)
Extra Trees (Track-A baseline)	0.8657	4.93	14.93
Slack Formula	-4.6019	32.26	127.26
Physics-Informed Hybrid	0.5597	9.05	39.49

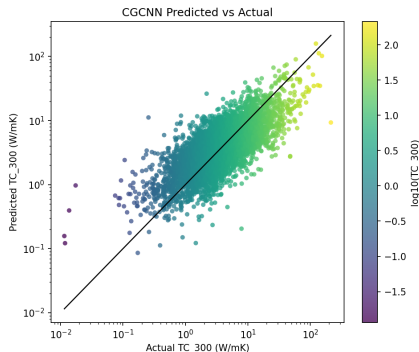


Interpretation: SHAP-vs-theory trend correlation: $r = 0.389$. Slack captures the leading Debye-temperature trend but misses material-specific scattering and structure effects.



CGCNN Crystal-Structure Model

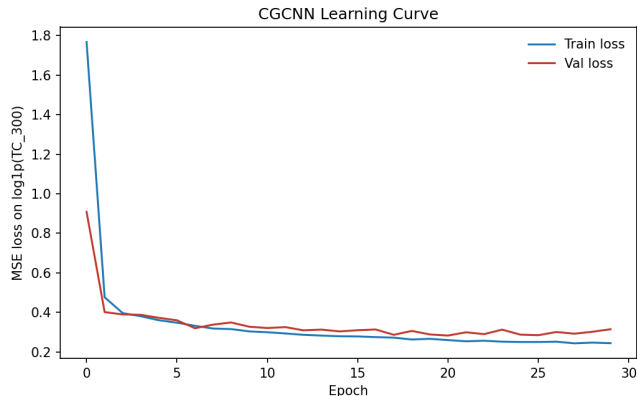
- MP matched structures: **4689 / 4753**
- CGCNN evaluated only on matched subset.
- CGCNN $R^2 = 0.4405$, RMSE = 7.64, MAPE = 77.75%.
- Results should not be inflated to the full 5,263-row dataset.



Interpretation: The graph model adds explicit crystal geometry, but its effective training size is limited by Materials Project coverage.



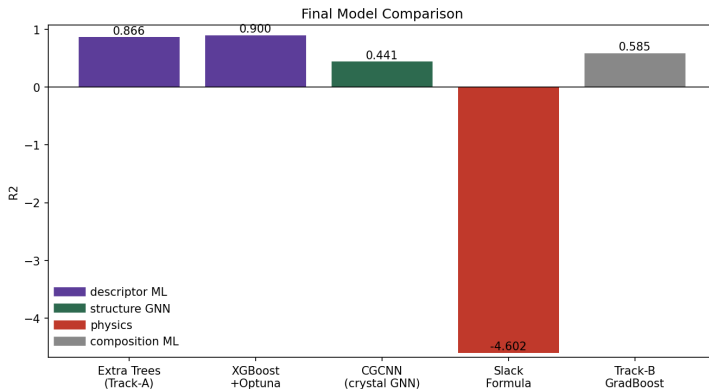
CGCNN Learning Curve



Interpretation: The learning curve reports mean train and validation loss across the SGN-grouped CGCNN folds.



Final Model Comparison



Interpretation: Descriptor ML remains strongest on the complete internal dataset; composition-only Track-B is limited, while physics and structure models clarify why.



Final Verified Status

- Track-A Extra Trees remains the primary complete-dataset baseline: $R^2 = 0.8657$, RMSE = 4.93, MAPE = 14.93%.
- XGBoost + Optuna reaches $R^2 = 0.8995$ under the requested SGN-grouped Track-A protocol.
- SHAP confirms DT_A_AGL is rank-1 by both mean absolute SHAP and permutation importance.
- Slack physics comparison gives SHAP-theory correlation $r = 0.389$.
- CGCNN is reported only on 4689 matched structures, not the full dataset.



References (Data and Methods)

- Matminer package docs: <https://hackingmaterials.lbl.gov/matminer/>
- Matminer composition featurizers (ElementProperty/Magpie):
<https://hackingmaterials.lbl.gov/matminer/matminer.featurizers.composition.html>
- Ward et al., npj Comput. Mater. (2016) composition descriptor framework.
- Ward et al., Comput. Mater. Sci. (2018) Matminer toolkit.
- Scikit-learn grouped CV, model selection, feature selection, and inspection docs.



Thank You

